Task 2

1. Why is this notation good/bad?

Scientific notation is useful because it makes very small and very large numbers easier to read. However, using only six decimal places causes some loss of precision.

2. Why do we see 1.234568 and not 1.234567?

Because the original number is 0.123456789. When it is written in scientific notation with six decimal places, the last digit is rounded up from 7 to 8.

Task 3

The NI USB-6211 has 16 single-ended or 8 differential analog input channels, 2 analog outputs, 4 digital inputs, 4 digital outputs, and two 32-bit counters. It uses a 16-bit ADC and supports a maximum sampling rate of 250 kS/s. The analog input ranges are ±0.2 V, ±1 V, ±5 V, and ±10 V.

How precisely can we measure a signal between 0 V and 5 V?

For a signal between 0 V and 5 V, we can use the ±5 V input range. With a 16-bit ADC, the resolution is:

Resolution = 10 V / 65536 = 0.000153 V ≈ 0.153 mV.

This is the theoretical ADC resolution. The actual measurement accuracy can be lower because of noise and other measurement errors.

Source: NI USB-6211 Specifications, National Instruments.

I tested the NI USB-6211 using NI MAX with On-Demand sampling and RSE configuration on channel Dev1/ai0. Since I did not have a switch, I manually changed the connection between AIGND and +5 V.

The measured voltage was approximately - 0.00044 V for the ground connection and 4.95 V for the +5 V connection. Both results were close to the expected values. The small differences and fluctuations may be caused by measurement noise and voltage offsets.

Task 4

I created a LabVIEW program to measure voltage using the NI USB-6211 DAQ device. The program allows the user to change the number of samples and sampling period from the Front Panel.

For testing, I used 100 samples with a sampling period of 100 ms. The voltage was measured on channel Dev1/ai0 using RSE configuration and On-Demand acquisition.

After the measurement, the data was displayed on a Waveform Graph and saved to a text file using Write Delimited Spreadsheet.vi with nine decimal places.

The measured voltage was close to 0 V, with small fluctuations caused by measurement noise and offset. The program worked as expected.